
MENU-DRIVEN STRING OPERATIONS USING FUNCTIONS IN C

Submitted By

Name : Rudra Pratap Singh
SAP ID : 590032811
Batch : 16

1. Problem Statement

Write a menu-driven C program that performs the following operations on a string, with each operation implemented in a separate function:

- Count the number of spaces, digits and special characters in a string.
- Count the frequency of a given character in a string.
- Toggle the case of each character in a string.
- Exit the program.

2. Objective

The objective of this project is to understand and apply the fundamental concepts of the C language: strings (character arrays), loops, decision making, the switch statement, user-defined functions and the standard library functions available in `<ctype.h>` and `<string.h>`. The program also shows how a menu-driven interface lets a user run several operations repeatedly without restarting the program.

3. Concepts Used

- **Character arrays (strings):** A string in C is an array of characters terminated by the null character `'\0'`. The program traverses the string until this terminator is found.
- **Functions:** Each operation is written as a separate function (`countTypes`, `charFrequency`, `toggleCase` and `readString`), which makes the program modular and easy to understand.
- **do-while loop and switch-case:** The do-while loop keeps displaying the menu until the user chooses Exit, while switch-case selects the operation for the entered choice.
- **<ctype.h> functions:** `isdigit()`, `isalpha()`, `isupper()`, `islower()`, `toupper()` and `tolower()` are used to classify and convert characters.
- **fgets() and strcspn():** `fgets()` reads a complete line including spaces (which `scanf("%s")` cannot do), and `strcspn()` removes the trailing newline character.

4. Description of Functions

Function	Purpose
<code>main()</code>	Displays the menu, accepts the user's choice, validates it and calls the required function using switch-case. Repeats until the user selects Exit.
<code>readString()</code>	Reads a line of text (including spaces) from the user into the string and removes the trailing newline.
<code>countTypes()</code>	Traverses the string and counts spaces, digits and special characters (any character that is not a letter, digit or space).

Function	Purpose
charFrequency ()	Accepts a character from the user and counts how many times it occurs in the string (case-sensitive).
toggleCase ()	Converts uppercase letters to lowercase and lowercase letters to uppercase; other characters remain unchanged.

5. Algorithm

5.1 Main Program (Menu)

1. Start.
2. Display the menu with options 1 to 4 and read the choice.
3. If the input is not a number, display an error message and show the menu again.
4. Using switch-case: for choice 1, 2 or 3 read the string and call the matching function; for 4 print the exit message; otherwise print "Invalid choice".
5. Repeat steps 2 to 4 until the choice is 4.
6. Stop.

5.2 Count Spaces, Digits and Special Characters

Initialise spaces = digits = special = 0. For each character of the string until '\0': if it is a space, increase spaces; else if it is a digit, increase digits; else if it is not an alphabet, increase special. Finally print the three counts.

5.3 Frequency of a Character

Read the character to be searched and initialise count = 0. For each character of the string, if it is equal to the given character, increase count. Finally print count.

5.4 Toggle Case

For each character of the string: if it is uppercase, convert it to lowercase; else if it is lowercase, convert it to uppercase; otherwise leave it unchanged. Finally print the modified string.

6. Source Code

File name: string_menu.c

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 200

/* Function prototypes */
void readString(char str[]);
void countTypes(char str[]);
void charFrequency(char str[]);
void toggleCase(char str[]);
```

```

int main()
{
    int choice;
    char str[MAX];

    do
    {
        printf("\n===== STRING OPERATIONS MENU =====\n");
        printf("1. Count spaces, digits and special characters\n");
        printf("2. Count frequency of a given character\n");
        printf("3. Toggle the case of each character\n");
        printf("4. Exit\n");
        printf("===== \n");
        printf("Enter your choice (1-4): ");

        if (scanf("%d", &choice) != 1)
        {
            printf("Invalid input! Please enter a number.\n");
            while (getchar() != '\n'); /* clear input buffer */
            choice = 0;
            continue;
        }
        while (getchar() != '\n'); /* remove leftover newline */

        switch (choice)
        {
            case 1:
                readString(str);
                countTypes(str);
                break;
            case 2:
                readString(str);
                charFrequency(str);
                break;
            case 3:
                readString(str);
                toggleCase(str);
                break;
            case 4:
                printf("Exiting the program. Thank you!\n");
                break;
            default:
                printf("Invalid choice! Please enter 1 to 4.\n");
        }
    } while (choice != 4);

    return 0;
}

/* Reads a line of text (including spaces) from the user */
void readString(char str[])
{
    printf("Enter a string: ");
    fgets(str, MAX, stdin);
    str[strcspn(str, "\n")] = '\0'; /* remove trailing newline */
}

/* Operation 1: counts spaces, digits and special characters */
void countTypes(char str[])
{
    int i, spaces = 0, digits = 0, special = 0;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ' ')

```

```

        spaces++;
    else if (isdigit((unsigned char)str[i]))
        digits++;
    else if (!isalpha((unsigned char)str[i]))
        special++;
}

printf("\nNumber of spaces          : %d\n", spaces);
printf("Number of digits           : %d\n", digits);
printf("Number of special characters : %d\n", special);
}

/* Operation 2: counts how many times a given character occurs */
void charFrequency(char str[])
{
    int i, count = 0;
    char ch;

    printf("Enter the character to search: ");
    scanf(" %c", &ch);
    while (getchar() != '\n');

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
            count++;
    }

    printf("\nThe character '%c' appears %d time(s) in the string.\n", ch, count);
}

/* Operation 3: toggles the case of every alphabet character */
void toggleCase(char str[])
{
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (isupper((unsigned char)str[i]))
            str[i] = tolower((unsigned char)str[i]);
        else if (islower((unsigned char)str[i]))
            str[i] = toupper((unsigned char)str[i]);
    }

    printf("\nString after toggling case: %s\n", str);
}

```

7. Sample Output

The following screen shows the program being run. Text typed by the user is shown in bold blue; the menu is displayed in full the first time and is shown again after every operation.

```
===== STRING OPERATIONS MENU =====
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
=====
Enter your choice (1-4): 1
Enter a string: Hello World 123 @$!

Number of spaces           : 3
Number of digits           : 3
Number of special characters : 4

[ menu is displayed again ]
Enter your choice (1-4): 2
Enter a string: programming is fun
Enter the character to search: g

The character 'g' appears 2 time(s) in the string.

[ menu is displayed again ]
Enter your choice (1-4): 3
Enter a string: Hello World 2026!

String after toggling case: hELLO wORLD 2026!

[ menu is displayed again ]
Enter your choice (1-4): 9
Invalid choice! Please enter 1 to 4.

[ menu is displayed again ]
Enter your choice (1-4): abc
Invalid input! Please enter a number.

[ menu is displayed again ]
Enter your choice (1-4): 4
Exiting the program. Thank you!
```

8. Test Cases

The program was compiled with gcc using the -Wall -Wextra options (no warnings) and tested with the following inputs:

No.	Operation	Input	Expected Output	Result
1	Count types	Hello World 123 @#\$!	Spaces: 3, Digits: 3, Special: 4	Pass
2	Count types	C Programming	Spaces: 1, Digits: 0, Special: 0	Pass
3	Count types	a1 b2 c3 !!	Spaces: 3, Digits: 3, Special: 2	Pass
4	Count types	(empty string)	Spaces: 0, Digits: 0, Special: 0	Pass
5	Frequency	programming is fun, ch = g	'g' appears 2 time(s)	Pass
6	Frequency	Banana, ch = a	'a' appears 3 time(s)	Pass
7	Frequency	Banana, ch = A	'A' appears 0 time(s)	Pass
8	Toggle case	Hello World 2026!	hELLO wORLD 2026!	Pass
9	Toggle case	C Language 101	c LANGUAGE 101	Pass

Note: the frequency operation is case-sensitive, so searching for 'A' in "Banana" gives 0 while searching for 'a' gives 3.

9. Conclusion

The program successfully performs all the required operations through a simple menu-driven interface. Splitting the work into separate functions makes the code organised, reusable and easy to debug. The project helped in understanding string handling, loops, conditional statements, switch-case, user-defined functions and the use of standard library functions in C. The program also handles invalid menu choices and non-numeric input gracefully.